FACULTY VIVA: FD E-COMMERCE ANALYTICS DASHBOARD

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1. A 60-second introduction you can learn

"Good morning. My project is the FD E-Commerce Analytics Dashboard, a business intelligence application that converts product, customer, order and marketing information into strategic decision support. The three main objectives are sales trend analysis, customer preference analysis, and campaign ROI evaluation. The application is built in Python with Streamlit and Plotly, supported by MongoDB for persistent storage. I also provide SQLite as an offline demo option. Six related datasets are validated and integrated before being analyzed. We can use interactive date, region and category filters, export reports and inspect customer RFM segments, cohort retention and campaign attribution. The included numbers are clearly marked as synthetic realistic demo data until I receive authorized records from a genuine store. I have automated tests for financial reconciliation and data integrity."

2. The 7-minute live demonstration

0:00–0:45 — Problem and project goal
Show the main banner. Explain the interdisciplinary link: software engineering + e-commerce operations + marketing data interpretation.

0:45–1:30 — Database and provenance
Open sidebar database label and Data Studio; show 6 entities and their record counts. If MongoDB mode is on, show the same collections in MongoDB Compass. If not, say this is the SQLite offline mode.

1:30–2:30 — Interactive sales
Change date and region filters; review delivered revenue, order count, AOV and trend charts. Explain why cancelled and returned statuses do not count.

2:30–3:40 — Customer preferences
Open Customer Intelligence. Explain recency, frequency, monetary values, returning customer share and cohort activity heatmap.

3:40–4:50 — Marketing performance
Open Campaign & ROI. Show spend, attributed revenue, ROAS, CPA and contribution ROI proxy. State why one campaign can be assigned to each order and why this cannot establish causal profit.

4:50–5:50 — Real interaction
In Data Studio, preview validation rules, empty Excel template and backup tool. If available, demo an additional seed or import only after backing up the previous dataset.

5:50–7:00 — Conclusions
Open Decision Lab, explain an insight with a concrete top-ranking product or channel, review tests and describe future genuine store API integration.

3. Faculty questions with precise answers

Q1. What is the exact aim of your project?
To integrate FD e-commerce sales, customer and marketing records to compute interpretable KPIs that support product, retention and advertising decisions.

Q2. Is this a shopping website?
No. It is a management and analytics dashboard *for* an e-commerce store. Payments, cart and checkout are beyond its present scope.

Q3. Why Python?
Python is effective for data cleaning, data validation, statistical summaries, metric testing, and connecting database records to interactive visualizations.

Q4. Why Streamlit?
It provides browser-based UI elements for Python computations without building a separate frontend/backend for a student analytics prototype.

Q5. Why MongoDB?
MongoDB persists the six logical datasets as collections. Each document has a batch identifier, and metadata selects the active consistent dataset. It is appropriate for flexible imported records, though strict cross-collection checks are performed in Python.

Q6. Why does your project also have SQLite?
It is a portable offline demonstration fallback. MongoDB is used when configured and running; SQLite must not be described as MongoDB.

Q7. Your synopsis says MERN. Isn't that a contradiction?
Yes, the first proposal used MERN, but the evaluated Phase 1 stack was Python/MongoDB/Streamlit. The implemented system follows the evaluated stack; the report explicitly documents the change instead of inaccurately claiming React or Express is in the final implementation.

Q8. Is your included data real?
No. It is explicitly labeled synthetic realistic test data generated by a seeded script. Actual operational analysis requires authorized source exports. I implemented data validation and Excel/CSV ingestion precisely to support replacing demo data with genuine records later.

Q9. What does dynamic data mean here?
The dashboard recomputes visualizations when filters or the active database dataset change. It does not yet contain a live store API or payment webhook.

Q10. Explain the ER model.
One customer may have many orders. One order has multiple items. Each item points to one product. A marketing campaign may be attributed to many delivered orders, but an order has at most one attributed campaign in the current single-touch model.

Q11. What is a primary key?
A field uniquely identifying a table's records, such as order_id. Foreign keys such as order_items.order_id refer to records in other collections, and their consistency is checked before data imports.

Q12. Explain revenue.
The sum of quantity × recorded item selling price, counted only for Delivered orders. Returned, cancelled and shipped orders are excluded from realized sales.

Q13. Explain AOV.
Average Order Value equals realized revenue divided by the number of unique qualifying delivered orders. If category filters are active, revenue is drawn only from matching order lines.

Q14. Explain RFM.
R=days since the most recent delivered order; F=number of delivered orders; M=total delivered spend. RFM segments are useful for retention analysis but are not a prediction model.

Q15. What is cohort retention?
A cohort groups buyers by first delivered purchase month. Retention at month k is the fraction purchasing in month k after their first-purchase month.

Q16. What is K-Means?
Unsupervised clustering groups customers based on standardized recency, frequency and monetary features, using a preselected cluster count. Cluster labels are numerical and not commercially validated personas by themselves.

Q17. Explain marketing ROAS.
Return on Ad Spend = campaign-attributed delivered revenue / associated campaign spend. A 3× ROAS means three rupees of attributed revenue per rupee of spend, not three rupees of profit.

Q18. Explain CPA.
Our CPA = allocated spend / count of attributed delivered orders. It is not necessarily CAC, because one buyer may have purchased before.

Q19. Explain ROI proxy vs actual profit.
The ROI proxy subtracts estimated product costs from attributed revenue before advertising spend. It excludes shipping, fees, taxes, overhead, full returns and incremental-causality testing, so it is not audited financial ROI.

Q20. How are campaigns matched to orders?
The conversions table contains documented campaign_id/order_id/customer_id associations; order dates are validated to fall inside campaign activity, and only delivered orders count. The demo generator simulates these associations.

Q21. What is data cleaning?
Checking required fields, parsing dates and numbers, removing or rejecting inconsistencies, validating IDs and reconciling transaction totals before loading the database.

Q22. What if an import is invalid?
Validation fails before replacement. An error is shown, leaving the last active dataset intact. A backup can be downloaded before any valid replacement.

Q23. How are you protecting customers?
Demo data contain invented names; the application supports a basic administrator gate for sensitive exports. Genuine customer records require stronger access control and anonymization before production use.

Q24. Can this website be deployed publicly?
A Streamlit frontend can technically be hosted and MongoDB can use an external cluster, but no public deployment is included here. Enterprise authentication and privacy controls must be implemented before hosting identifiable customer data.

Q25. How did you test it?
Ten automated tests cover deterministic data, order reconciliation, status rules, category scope, marketing aggregation, prorated costs, RFM/cohorts, invalid imports and database round-trips. I can run `python -m pytest tests -q` to prove it.

Q26. What are the project's main limitations?
Synthetic rather than verified business data, no live platform integration, single-touch marketing attribution, assumed uniform spend allocation, incomplete cost accounting, and limited administrator authentication.

Q27. Why is this interdisciplinary?
Computer applications manages data architecture, programming and UI; e-commerce and marketing provide sales, segmentation, attribution and financial KPI interpretation.

Q28. What strategic insights can a store manager obtain?
Which product category currently generates the most delivered sales, which historical customers appear dormant, which channel has stronger observed ROAS, and which regions show stronger demand, provided reliable source data support those results.

Q29. Explain the difference between operational and analytical databases.
An operational database stores granular records needed to run a business; an analytical layer derives summaries

and trends. This prototype stores granular imported records and computes analytical results in Pandas, rather than implementing a separate enterprise data warehouse.

Q30. How would you add genuine real-time data?
Obtain an approved Shopify/WooCommerce/other order-platform API or webhook, authenticate securely, validate events, handle duplicate deliveries and failures, store timestamped records and refresh cached analytics. It is a future feature, not a claim about this prototype.

4. Things faculty may specifically challenge

- If asked why the original synopsis mentions React: answer using Q7 and revise the document.
- If asked whether fake names are genuine customers: say **no** instantly, then explain the provenance and collection plan.
- If asked whether a saved MongoDB database is real time: explain the difference between persistence, dynamic UI recomputation, and live third-party ingestion.
- If asked whether high ROAS proves successful paid ads: no, attribution may be biased; ROI requires cost assumptions and causal measurement.
- If asked whether static stock indicates inventory accuracy: no; it is a demonstration snapshot and not an inventory ledger.

5. Key commands to memorize

```
# Start app after dependencies installed
python -m streamlit run app.py

# Load transparent synthetic demo
python seed.py --seed 42 --customers 1250 --orders 5200

# Run automated tests
python -m pytest tests -q

# Create virtual environment on Windows
py -3 -m venv .venv

# Install declared packages
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
```

6. Statement for the end of your presentation

"The key contribution of this project is not just drawing charts; it is validating and integrating records from six related entities, defining the business meaning of each KPI, and showing how those indicators support better decisions. My working prototype is dynamic, testable and ready to be populated with authorized genuine store exports, while transparently identifying its current synthetic dataset and financial limitations."